

Columbia Tap OptiFlow

Optimal implementation of wet bioswale check dams along Houston's Columbia Tap Rail Trail

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Background - Bioswales

- **Green infrastructure** that captures runoff
- Slows runoff and cleanses water
- Water pools behind check dams
- Pollutants settle out via sedimentation
- Wetland plants absorb nutrients and filter pollutants, clean water slowly reabsorbs

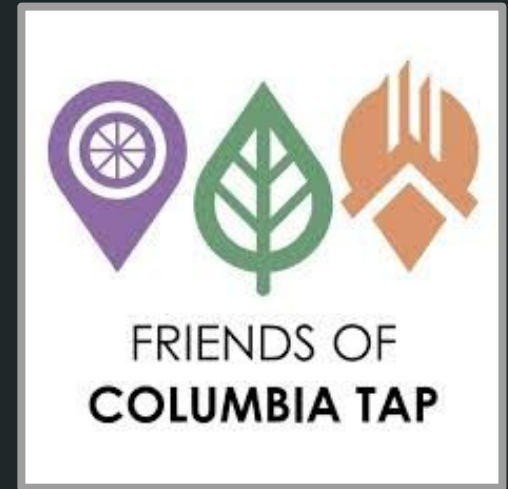
Benefits:

- **Pollutant** filtration
- **Cooling** along trail
- **Air quality** improvement



Columbia Tap Trail

- 4-mile rail-trail through Third Ward and EaDo
- Passes Texas Southern University, connects to Brays Bayou Greenway and Hermann Park
- Friends Of Columbia Tap
 - Community protecting the Columbia Tap Trail, within the Historic Third Ward community.
 - **Key community partner**



Bioswales already present on trail (Rayaan)

- Resident-implemented wet bioswales
- AIDS Memorial Garden visit
- One of three conducted trail visits



Goal of this project (Rayaan)

- **Optimize placement of dams** along existing drainage ditches
 - Simple installation of forebay or dam can transform ditch segment into wet-bioswale
- Maximize water treatment in a 1-year storm event while minimizing habitat damage and unnecessary pooling
- Conduct site visits to identify and collect relevant trail parameters
- We put this trail data to work in a mixed integer linear program

Mixed Integer Program - Setup

We break trail corridor ditch into segments of length l

- S = set of all ditch segments
- $A = \{(i, j) \text{ for } i, j \text{ are adjacent segments with } j \text{ on inlet side of drainage ditch (segment } i \text{ flows into segment } j)\}$
- r_i = is total rainfall catchment by segment i during 2-year storm (constant)
- c_i = capacity of ditch segment i (volume of ditch calculated with height = dam height)

h_i = heuristic habitat value of segment i (0-1 scale)

- $L_{ui} = \{j : j \text{ leads into same inlet and lies within 5 segments (75 ft) uphill (not necessarily adjacent) of segment } i\}$
- $L_{di} = \{j : j \text{ leads into same inlet and lies within 5 segments (75 ft) downhill (not necessarily adjacent) of segment } i\}$

Integer Program - Decision Variables

$$x_i = \begin{cases} 1 & \text{if dam installed at downhill side of segment } i \\ 0 & \text{otherwise} \end{cases}$$

$$y_i = \begin{cases} 1 & \text{if forebay installed on uphill side of segment } i \\ 0 & \text{otherwise} \end{cases}$$

$$u_i = \begin{cases} 1 & \text{if bioswale exists on segment } i \\ 0 & \text{otherwise} \end{cases}$$

z_i = water flow(volume) during rain event into segment i

v_i = water volume captured and treated by segment i

w_i = water volume captured and not treated by segment i

Objective:

$$\max \sum_{i \in S} (v_i + \underbrace{ah_i u_i}_{\text{Bioswale Suitability Term}} - \underbrace{bw_i}_{\text{Unnecessary Pooling Term}})$$

Water
Treated by
Bioswales

Bioswale
Suitability
Term

Unnecessary
Pooling
Term

Habitat
Preservation
Weight

Pooling
Penalty
Weight

Integer Program - Implementation Constraints

- T_d and T_f are parameters we give to our model to determine best implementation plan given limited resources
- With these 4 parameters (T_u , T_d , a , b) we can construct implementation plans for various circumstances and preferences

$$\sum_{i \in S} x_i \leq T_d$$

Where T_d is number of dams to install (test various values). Limit number of dams installed

$$\sum_{i \in S} y_i \leq T_f$$

Where T_f is number of forebays to install (test various values). Limit number of forebays installed

Integer Program - Flow Constraints

$$z_j = r_j + z_i - v_i - w_i$$

$$\forall (i, j) \in A$$

Flow constraint for segments in S_1

$$z_i = r_i$$

$$\forall i \in S_2$$

Flow constraint for segments in S_2 (no segment predecessors)

$$w_i + v_i \geq z_i - M(1 - x_i)$$

Sufficiently large M in this case: $M = r_i + \sum_{j \in L_{ui}} r_{uj}$

$$\forall i \in S$$

On segments dammed below that are not bioswales, water is gathered that is not properly treated

Integer Program - Functional Constraints

$$u_i \leq x_i + \sum_{j \in L_d i} x_j$$

$$\forall i \in S$$

Segment must be dammed below to be considered valid bioswale

$$u_i \leq \sum_{j \in L_u i} (y_j + x_j) + y_i$$

$$\forall i \in S$$

Segment requires above forebay or check dam to be considered valid bioswale

$$v_i \leq z_i$$

$$\forall i \in S$$

Treated water cannot exceed inflow to segment

$$v_i \leq u_i c_i$$

$$\forall i \in S$$

Segment must be valid bioswale to treat water, and water treated cannot exceed capacity of segment

Soil Analysis

Soil Type

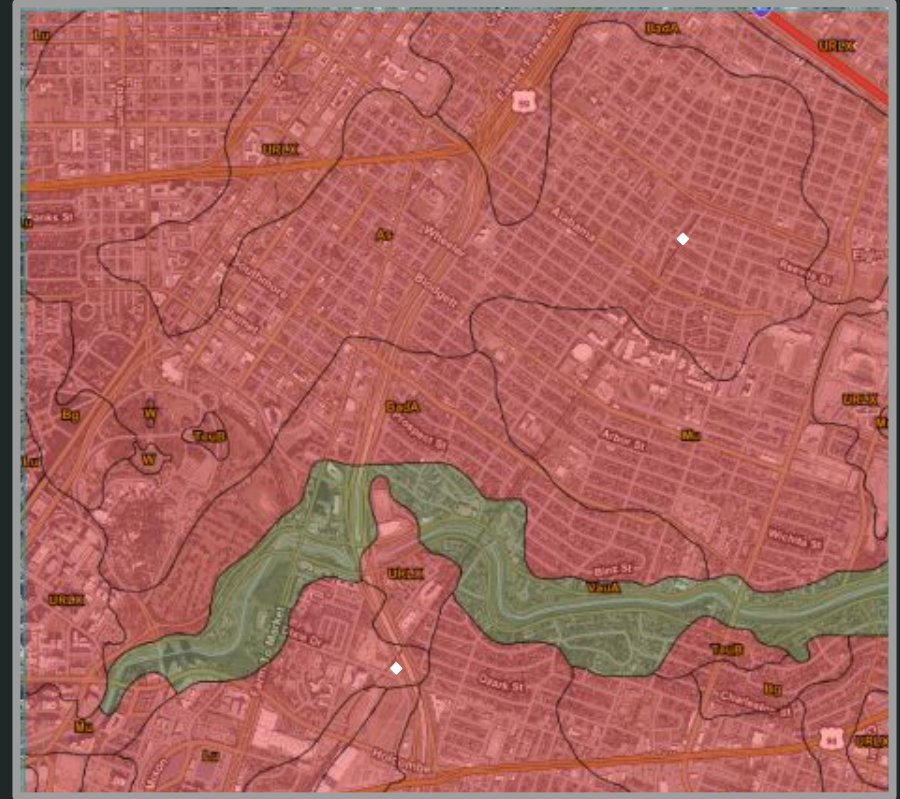
- Red Area: Type D
- Gray Area: Type C/D
 - Not considered due to no inlets

Implications

- Slow water infiltration
- **Wet** bioswales

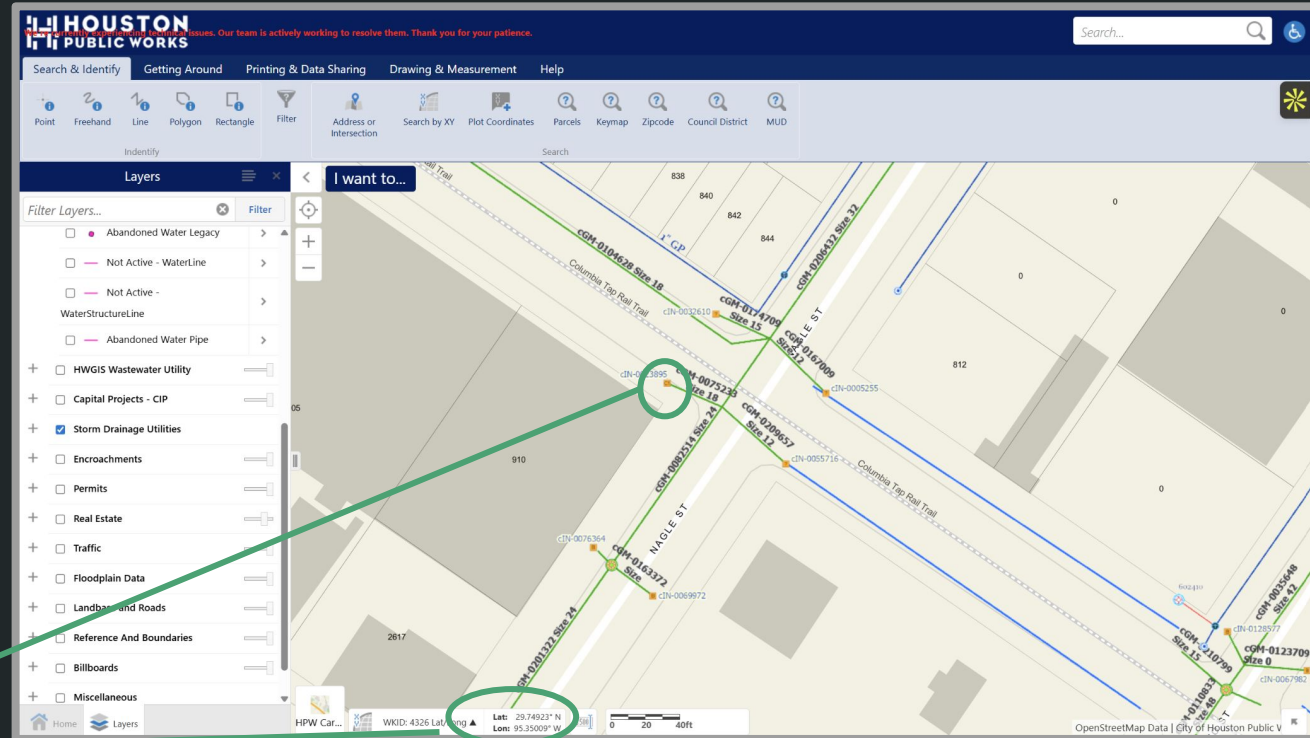
Source: Web Soil Survey

<https://websoilsurvey.nrcs.usda.gov/app/WebSoilSurvey.aspx>



Utilities and inlet location (Rayaan)

- Found all inlets and utilities along the trail
- Avoided sewage utilities
- Visited these coordinates for verification



Key information

Catchment areas (Jonathan)

Minimum Area Required

- Given
 - Ditch width: 8 ft
 - Check dam depth: 1.5 ft
 - 1 year storm event 24 hour total rainfall: 0.319 ft
- Variable
 - Segment length: 15 ft
 - Runoff coefficient: 0.6 (avg. residential and commercial areas)
- Calculations
 - $.191 = \text{total rainfall} * \text{Runoff coefficient}$
 - $180 = \text{width} * \text{depth} * \text{length}$
 - $180 / .191 = 942.41 \sim 30.70 \times 30.70$

Sources:

https://hdsc.nws.noaa.gov/pfds/pfds_map_cont.html?bkmrk=tx

Methodology

- Data
 - 2018 Harris County LiDAR
- Technique
 - Convert data to Digital Elevation Models
 - Evaluate catchment areas using ArcGIS
- Results
 - Previously identified possible locations all catchment areas > 1000 feet²
 - Not a limiting factor

Sources: <https://data.tnris.org/>

Habitat Suitability Score for Potential Bioswales

Rubric:

- Focused on 25 segments for each potential bioswale, each 15 feet long
- Each trail segment is scored from 0 to 1 based on how well it can support a functioning bioswale

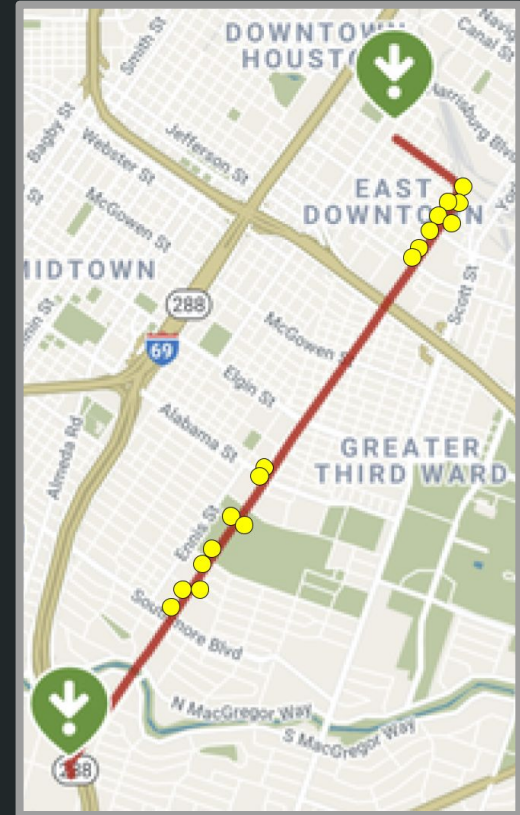
Score	Conditions
0.0 - 0.2	Unsuitable: dense forest, steep slopes, or utility conflicts
0.3 - 0.4	Poor: heavy tree cover, tight space, or drainage issues
0.5 - 0.6	Moderate: mixed conditions, slight vegetation hinderance
0.7 - 0.8	Good: open land, adequate space, gentle slope
0.9 - 1.0	Ideal: ample space, ideal slope between 0.5% - 2%

Habitat Suitability Visit

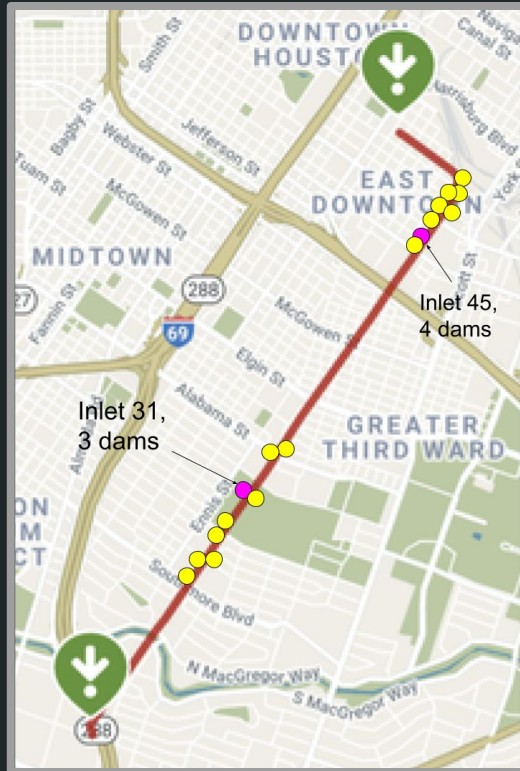


Results - Parameter Effects on Dam Placement

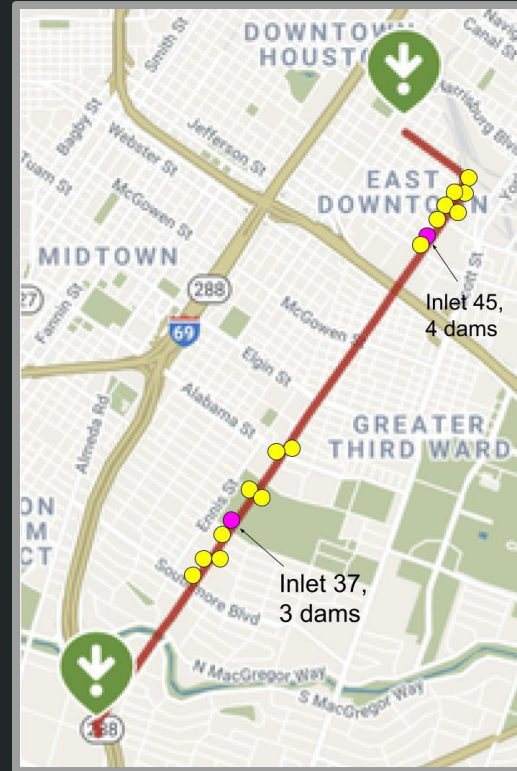
- First, we will fix T_d (quantity of dams) to be 7
- We will first test various values for a (habitat suitability weight) and b (unnecessary pooling weight)



Fix $b = 0.1 \rightarrow$ test two values of a

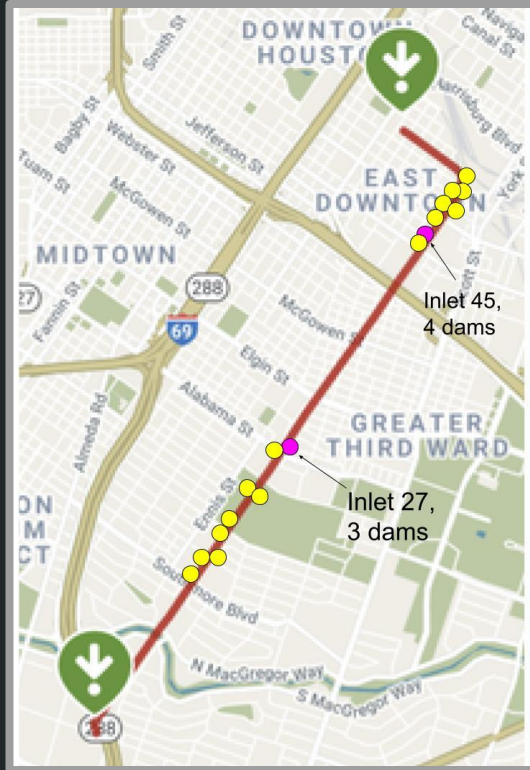


- $a = 0$
- 4500 cf

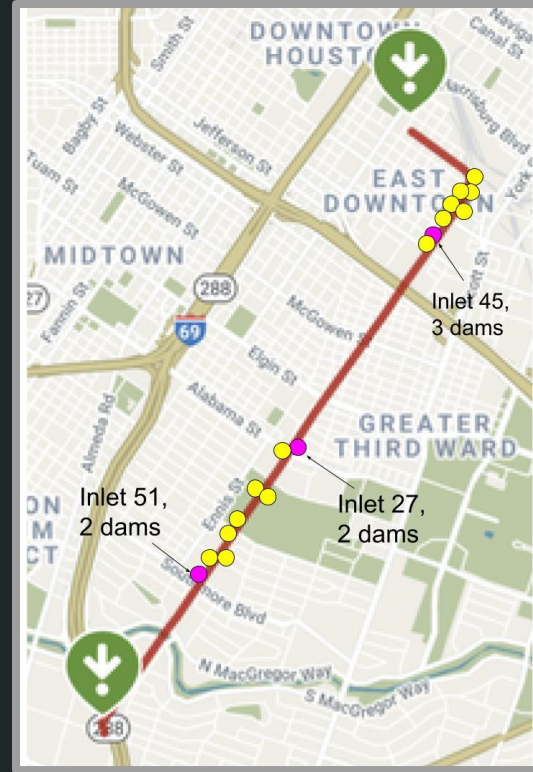


- $a = 250$
- 4320 cf

Fix $a = 100 \rightarrow$ test two values of b



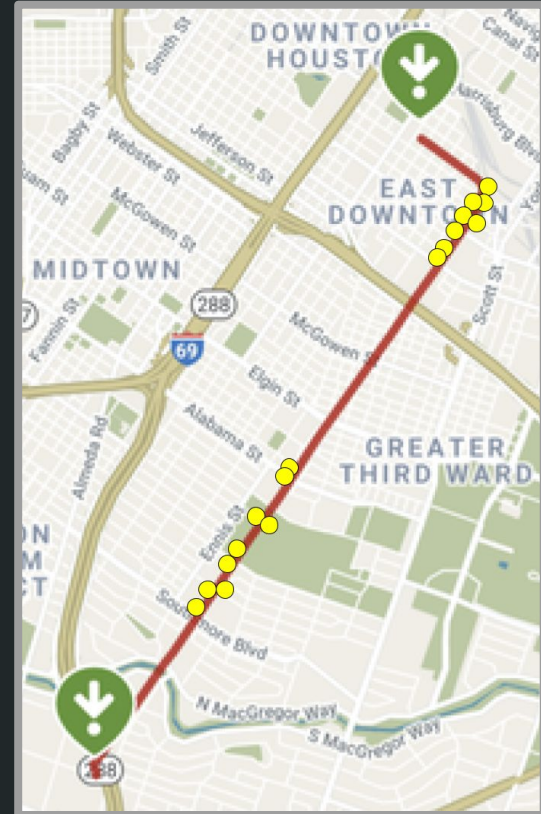
- $b = 0.05$
- Water catchment: 4500 cf



- $b = 0.5$
- Water catchment: 3060 cf

Damming Locations with Quantity of Dams to Install

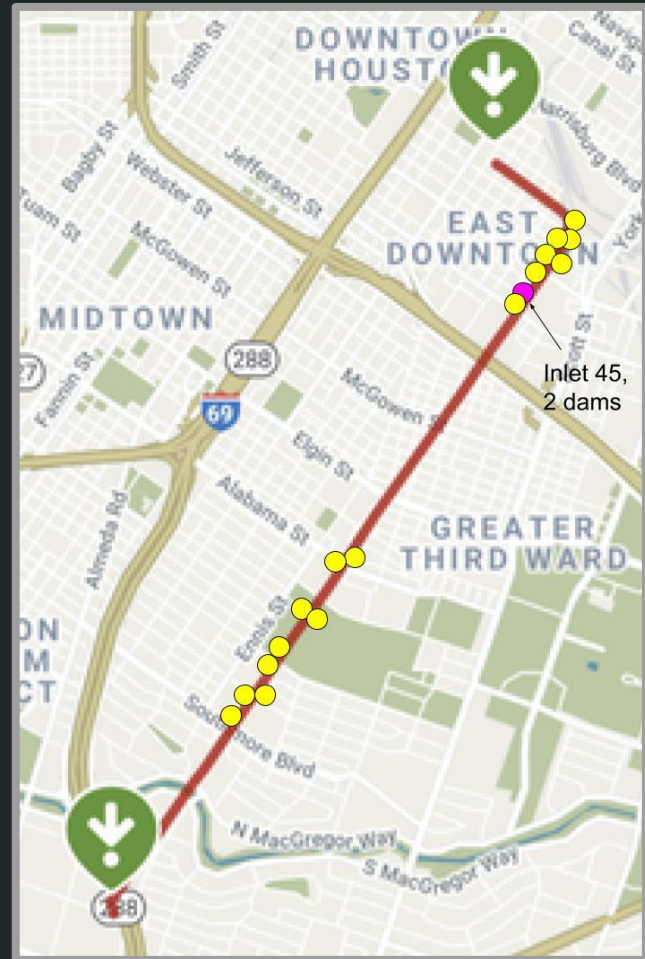
- For each test, we will fix $a = 100$ and $b = 0.1$
- We will test various values of T_d (number of dams to install)



Number of Dams to Install - 2

By setting $T_d = 2$, the damming location results yield:

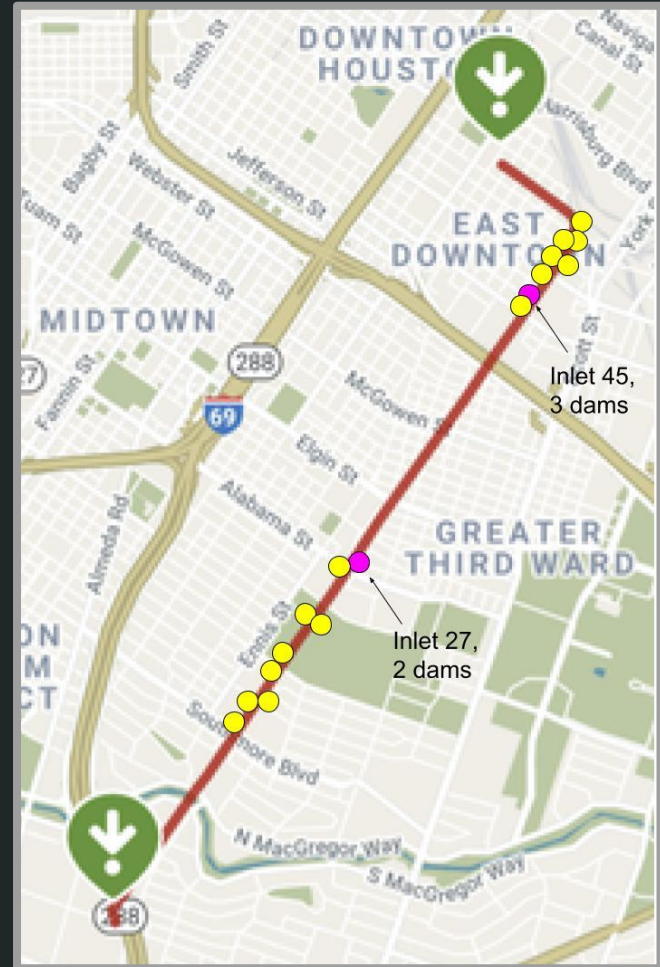
- 165 and 240 feet away from inlet 45
- Water catchment: 900 CF



Number of Dams to Install - 5

By setting $T_d = 5$, the damming location results yield:

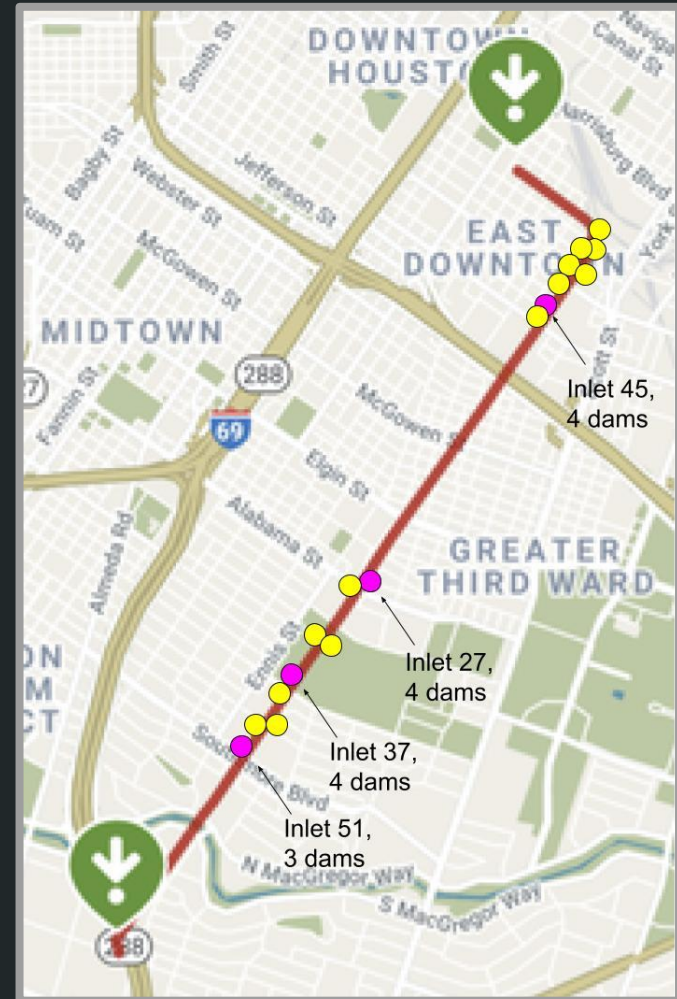
- 135 and 210 feet away from inlet 27
- 90, 165, and 240 feet away from inlet 45
- Water catchment: 2700 CF



Number of Dams to Install - 15

By setting $T_d = 15$, the damming location results yield:

- 0, 75, 150 and 210 feet away from inlet 27
- 0, 60, 135, and 210 feet away from inlet 37
- 15, 90, 165, and 240 feet away from inlet 45
- 60, 135, and 210 feet away from inlet 51
- Water Catchment: 9540 CF



Conclusions - colt